

Higher Criticism Adaptive Retrieval on Amazon Queries

From a Controlled Proof-of-Concept to a Calibration Breakthrough

1 Motivation: Controlled Environments for HC

Higher Criticism (HC) replaces fixed top- k retrieval by *statistically* deciding how many documents to return per query: cosine similarities are turned into p-values against a null distribution, and the HC statistic locates the threshold where relevant documents stop concentrating. The aim of this effort was not merely to benchmark HC, but to build **controlled environments** in which its behaviour can be *understood*. We wanted datasets that (i) emulate a realistic retrieval setting (Amazon product search), (ii) have fully known ground-truth structure — every document’s relevance tier and every query’s true K are fixed by construction — and (iii) expose a single, tunable difficulty knob. With these three properties we can isolate the conditions under which HC works correctly and the conditions under which it fails, and attribute each failure to a specific mechanism rather than guess at it. The work follows exactly this axis: a dataset engineered to *fit* HC’s core assumption, then harder datasets that *controllably violate* it, and finally a diagnostic study that turns the known structure into a precise account of failure — and a fix.

2 Proof of Concept: the category-key Dataset

The **category_key** dataset is the regime built to satisfy HC’s assumptions. Each of 44 queries asks “What are the largest cities in [country]?”; the corpus holds 726 documents (506 relevant city passages, 220 distractors). The number of relevant documents varies by query — small ($K = 4-5$), medium ($K = 9-11$), and large ($K = 18-24$) buckets — so no single fixed k can serve all queries, making adaptivity necessary by design. Crucially, relevant passages genuinely cluster at high cosine similarity (BGE-base-en-v1.5 embeddings), and a *per-query* empirical null is near-perfectly calibrated (pooled KS = 0.003, 44/44 queries pass). This is the environment in which HC *should* work — and it does:

Method	Recall	Precision	F1
Top-10 (best fixed- k)	0.800	0.766	0.783
Top-20	0.939	0.511	0.662
HC (mean $k = 9.7$)	0.852	0.938	0.893

Table 1: **category_key**: HC beats every fixed- k baseline on F1 by adapting k per query (true- K correlation 0.835). This validates the HC machinery and gives a clean reference point.

3 Turning the Difficulty Knob: Compound Queries

To stress HC in a *controlled* way, we introduced **compound queries** that combine a category filter with a numerical price constraint, e.g. “Find Electronics products priced above \$150.” This is a deliberate, measurable violation of HC’s premise: standard text embeddings match category semantics but **cannot reason about numerical inequalities**, so hard negatives (right category, wrong price) sit right next to relevant documents in cosine space. The difficulty is a single, well-defined knob, and every document’s tier is known by construction.

Two corpora instantiate this regime (OpenAI text-embedding-3-small, FAISS IndexFlatIP): **amazon_compound** — 80,000 documents across three departments, 135 queries, only 4.8% relevant — and the larger **electronics_100k** — 100,000 single-department documents, 44 queries, 1.2% relevant, with 44.5% *uncapped* hard negatives. To scale, we also adopted the architectural constraint of a **single global**

null (one calibrated distribution shared by all queries at inference), replacing `category_key`’s per-query nulls.

The failure. Under the calibrated global null, HC breaks on `amazon_compound`: **55 of 135 queries produce a negative HC statistic and return $k=0$** — nothing at all. The controlled design is what lets us ask *which* 55, and *why*.

4 Diagnosing the Failure: Three Mechanisms

Because relevance and true K are known, the 135 queries can be partitioned exactly and each failure attributed to a mechanism:

1. **Embedding limitation (24 “irreducible” queries).** For these the price signal is simply absent from the cosine: relevant documents are scattered through the ranks (similarity gap ≈ 0), below the Donoho–Jin detection boundary. *No* statistical test can recover them — the pipeline failed before HC ran.
2. **Calibration mismatch (31 “fixable” queries).** Here the embedding *does* separate, but the per-query non-relevant z-score spread ($\sigma_q \approx 1.74$) is much narrower than the global null ($\sigma_{\text{null}} \approx 2.06$). This inflates p-values and trips the negative-HC gate. The std-ratio mismatch is the *dominant* predictor of HC failure ($r=0.83$ with the HC statistic).
3. **Signal dilution (general).** With a fixed scan window γ , the HC win condition $p_{(i)} < i/N$ tightens linearly as the pool N grows, so a bigger pool is not a free fix — it trades calibration gains for dilution losses.

The key realization: the 31 fixable queries are broken by *miscalibration*, not by missing signal — and the miscalibration is a direct consequence of the global-null constraint we adopted for scale. The fix must therefore *recalibrate per query without abandoning the single global null*.

5 The Breakthrough: `rescale_trim`

`rescale_trim` is a label-free, per-query transform applied to the z-scores *before* the global-null lookup. For each query we estimate a robust noise scale $\hat{\sigma}_q$ as the standard deviation of the pool z-scores *after dropping the top γ ranks* (where relevant documents concentrate and would otherwise contaminate the estimate), then rescale:

$$z'_i = z_i \cdot \frac{\sigma_{\text{null}}}{\hat{\sigma}_q}, \quad \hat{\sigma}_q = \text{std}(z_{(\gamma+1:N)}).$$

This maps every query onto the same scale as the global null, so the empirical CDF lookup becomes meaningful, and it directly cancels the $r=0.83$ std-ratio mismatch identified above. The transform uses only observable pool statistics (no labels) and is applied uniformly to every query — the null *itself* stays a single global distribution, so the architectural constraint is preserved.

Group	n	Top-50	Global null (no rescale)	HC + <code>rescale_trim</code>
positive_hc	77	0.233	0.253	0.251
fixable	31	0.164	0.000	0.157
irreducible	24	0.014	0.000	0.012

Table 2: Per-group macro-F1 on `amazon_compound`. `rescale_trim` **recovers 30 of 31 fixable queries** ($0.000 \rightarrow 0.157$), preserves the 77 already-working queries (within 0.002), and correctly leaves the 24 embedding-limited queries untouched. The failure mode is substantially eliminated — without touching the global null and without using any labels.

6 Final Results and Generalization

Best-in-class retrieval. On `amazon_compound`, `HC + rescale_trim` is the *only* method whose F1 strictly beats every fixed top- k baseline, at an adaptive mean $k=34$. It also leads a panel of competing selection rules evaluated on the identical pipeline:

Method	Recall	Precision	F1	Mean k
Top-20 (fixed)	0.179	0.192	0.185	20.0
Top-50 (fixed)	0.287	0.139	0.187	50.0
HC + rescale_trim	0.209	0.209	0.209	34.0
Berk-Jones + rescale_trim	0.238	0.176	0.202	34.8
Adaptive- k (2025)	0.104	0.204	0.138	9.0
Cluster / CAR (2025)	0.598	0.060	0.110	416.5
Benjamini-Hochberg	0.064	0.087	0.074	1.6
Bonferroni	0.014	0.049	0.022	0.3

Table 3: Algorithm comparison at the canonical pool $N=1000$. HC posts the best F1 and highest precision; Berk-Jones follows closely. The FWER/FDR rules collapse (flat thresholds cannot find an adaptive cut-off); CAR reaches HC’s F1 only when its pool is hand-tuned to $N=100$, whereas HC needs no pool-size hyperparameter.

Generalization to scale. On the larger, single-department `electronics_100k` (100K docs, 44.5% uncapped hard negatives), HC again attains the best retrieval F1 (0.192 vs. 0.185 top-20, 0.166 top-50), +34.6% recall over top-20 at adaptive $k=34$. The advantage carries through the full RAG chain: against top-20, HC lifts **Token F1 +15.0%**, **Title F1 +19.6%**, and **Fuzzy-Title F1 +17.0%** (gpt-5-mini generation, gpt-5.2 judge), reaching 94–97% of top-50’s generation quality with 32% fewer documents — the adaptive- k benefit is not a small-dataset artifact.

Dataset	Regime	Docs / Queries	Best fixed- k F1	HC F1
<code>category_key</code>	clean semantic, var- K	726 / 44	0.783	0.893
<code>amazon_compound</code>	category+price, 3 depts	80K / 135	0.192	0.209
<code>electronics_100k</code>	category+price, 1 dept (100K)	100K / 44	0.185	0.192

Table 4: The journey at a glance. HC is the best method in every regime. The absolute F1 drop from `category_key` to the compound corpora reflects *dataset difficulty* (embeddings blind to price, $\leq 4.8\%$ relevant), not a degradation of HC — which remains the top performer throughout.

7 Conclusions

The controlled-environment strategy paid off. Engineering datasets with known structure and a single difficulty knob let us (1) confirm HC’s adaptive- k advantage where its assumptions hold (`category_key`), (2) *decompose* its failure on hard compound queries into embedding-limited, calibration-mismatched, and dilution components, and (3) design `rescale_trim` — a label-free per-query recalibration that eliminates the dominant (calibration) failure mode while respecting the single-global-null constraint. The outcome is the best retrieval F1 of any method tested, generalizing from 726 to 100,000 documents and through to end-to-end RAG quality. The residual “irreducible” failures are a property of the embeddings, not of HC, and point to the natural next step: hybrid retrieval that combines embeddings for category semantics with structured metadata filtering for the numeric constraint.